SungGeun **AN**

ssunggun2@snu.ac.kr | ssunggun2.github.io | linkedin.com/in/sunggeun-an | github.com/ssunggun2

RESEARCH INTEREST

- Whether multimodal language models reason from the *evidence in front of them* or from priors learned during pretraining.
- Benchmark and metric design that separates *perception* failures from *reasoning* failures.
- Evaluation methodology for language models, including pass@k diagnostics and human-baseline studies.
- **Keywords**: Multimodal LLMs, physical reasoning, evaluation and benchmark design, retrieval-augmented generation, vision-language models.

EDUCATION

03/2024 – 08/2026 **Seoul National University**
M.S. in Data Science
Graduate School of Data Science (GSDS), SKI-ML Lab
M.S. thesis, UNREAL: Uncovering Physics Reasoning Limits in MLLMs

2012 – 2021 **Seoul National University**
B.S. in Geography Education
College of Education, Department of Geography Education

EXPERIENCE

03/2024 – 08/2026 **M.S. Student, Research Assistant**
Seoul National University, SKI-ML Lab, Seoul, South Korea

- M.S. thesis (UNREAL): a benchmark that tests whether multimodal LLMs reason from the video in front of them or fall back on physical priors learned in pretraining. Synthesized 1,528 videos in Kubric under 15 categories of deliberate physics violations with staged QA, and proposed two metrics (Reality Bias Rate, World-Model Inconsistency Rate) validated against a 13-annotator human study
- Industry collaboration with LG AI Research (Feb-Aug 2026): owned dataset discovery and evaluation criteria on the data team, extending a Korean LLM (EXAONE-3.5-7.8B-Instruct) from finance QA into long-tail expert domains (circuit design and HDL, biomedicine) under a limited data budget. Designed a pass@k diagnostic framework (N=128) separating latent from trained capability, graded benchmarks into difficulty tiers by the pass@1 to pass@128 gap, and set the evaluation-set criteria after showing multiple-choice formats saturate

12/2021 – 12/2023

Data Scientist
Aimpact, Seoul, South Korea

- Arrange, a direct-to-consumer order processing platform whose core feature is parsing free-form Korean order messages from SMS and KakaoTalk into structured orders (KR patent 10-2287410). 6,150 merchants, 22,287 listed products, and KRW 28.6B in cumulative transactions as of Aug 2022.
- Built and trained Korean language models that extract address, recipient, product, and quantity fields from free-form order messages, handling misspellings, abbreviations, and mixed road-name and lot-number address formats
- Designed and deployed NLP APIs using FastAPI, serving real-time inference
- Implemented OCR pipelines integrating Naver Clova and GCP Vision APIs
- Migrated backend from Flask to FastAPI with async processing for 3x throughput
- Integrated Redis caching and AWS SQS/Lambda for scalable async workflows
- Deployed and managed services on AWS ElasticBeanstalk

PUBLICATIONS

1. **Sieve and Sage: Efficient Distraction Filtering for Reliable RALM Abstention**.
   In *Findings of EMNLP*, 2026.
   Jongbin Won, **Sung Geun An**, Jay-Yoon Lee.

SERVICE AND ACTIVITIES

03/2026 – 08/2026

Lab manager, SKI-ML Lab, Seoul National University
Ran lab operations: intern and member selection and onboarding, seat and resource allocation, and research budget handling with the advisor and administrative staff. Set the usage policy for the shared GPU servers and the LLM API gateway, arbitrating allocation between members and handling outages. Managed the weekly group meeting, paper seminars, and study groups, served as the lab's contact point for external collaborations and department events, and documented the procedures and infrastructure setup so the role could be handed over.

03/2026 – 08/2026

Organizing committee, SNU AI Challenge 2026
Designed the Kaggle task and evaluation metric; scored final reports against a rubric.

02/2026 – 08/2026

Steering committee member, GSDS graduate student council
Helped plan and run department events.

03/2025 – 06/2025

Teaching assistant, Machine Learning and Deep Learning 1, SNU GSDS
About 100 graduate students. Supported PyTorch implementations of CNN, RNN, and Vision Transformer models, debugged training and environment issues in office hours, and graded assignments and exams against a shared rubric.

09/2024 – 12/2024

Teaching assistant, Foundations of Mathematics and Statistics for Data Science, SNU GSDS
About 50 graduate students. Ran review sessions and office hours on linear algebra, multivariate calculus, and probability and statistical inference, and graded assignments and exams against a shared rubric.

08/2025 – 08/2025

Teaching assistant, Samsung Electronics DS2 program
NLP and LLM intensive for Samsung Electronics engineers, held at the SNU-Samsung joint research center. Ran TA office-hour sessions spanning the mathematical foundations of deep learning, word embeddings, RNN and LSTM, Transformer encoder-decoder, the Hugging Face stack, prompting and chain-of-thought, LangChain and API usage, retrievers and FAISS-based RAG, and alignment tuning (RLHF, DPO). Resolved environment and pipeline issues during labs and proctored evaluations.

08/2025 – 08/2025

Teaching assistant, Hyundai Motor Company AI training

AI and NLP training program for practicing engineers at Hyundai Motor Company. Ran hands-on office-hour sessions, resolved environment and pipeline issues during labs, and proctored evaluations.

SKILLS

Machine Learning & Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers
NLP & Multimodal	Large Language Models, Multimodal LLMs, Vision-Language Models, Retrieval-Augmented Generation, Named Entity Recognition, OCR
Evaluation & Benchmarking	Benchmark Design, Human Evaluation Study Design, pass@k Diagnostics, Prompting Strategy Analysis
Programming	Python, SQL
MLOps & Cloud	AWS (SQS, Lambda, ElasticBeanstalk), Docker, FastAPI, Redis
Languages	Korean (native speaker), English (professional working proficiency)